

REVIEW ARTICLE OPEN ACCESS

Smart Grid Management Enhanced by Large Language Models: From Fault Detection to Real-Time Optimization

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ABSTRACT

The rapid advancement in digitalization has led to significant innovations in smart grid systems, enhancing energy management, fault detection, and operational optimization. The integration of artificial intelligence (AI), particularly large language models (LLMs), plays a pivotal role in transforming smart grid functionalities. This study explores the potential of LLMs in optimizing real-time operations and improving fault detection and predictive maintenance in smart grids. LLMs, known for their ability to process vast amounts of unstructured textual data, can offer superior contextual understanding and reasoning, making them suitable for managing the complex, dynamic data produced by smart grids. This paper evaluates how LLMs can address the challenges posed by data heterogeneity, quality, and interpretability in fault detection systems. Furthermore, the research identifies the key challenges in integrating LLMs into smart grid systems, including scalability issues, security vulnerabilities, and model transparency. Through a detailed analysis of existing solutions, this paper outlines the current advancements in LLM-based models for smart grid optimization and fault detection, emphasizing their potential to enhance grid resilience, operational efficiency, and cybersecurity. Finally, this work highlights the need for further research to overcome the challenges and unlock the full potential of LLMs in smart grid management, providing a roadmap for future developments in AI-powered energy systems.

1 | Introduction

The digitalization of energy systems is undergoing rapid evolution, driven by the convergence of smart technologies, automation, and data analytics to enhance energy efficiency, reliability, and sustainability. Key trends include the deployment of artificial intelligence (AI), internet of things (IoT), and digital twins (DTs) in energy production, distribution, and consumption,

enabling intelligent energy management, fault detection, and peer-to-peer energy trading [1]. Studies reveal that digitalization significantly improves process efficiency, health, safety, and environmental performance in energy sectors through technologies such as smart grids, automation, and machine learning (ML) [2]. AI and ML are playing a transformative role in grid modernization by enhancing the resilience, efficiency, and intelligence of power systems. One major application is the

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development of self-healing grids that use AI techniques for automatic outage detection, diagnosis, and restoration, thus reducing downtime and human intervention [3]. ML models are also being employed to optimize real-time energy consumption and forecasting through attention mechanisms and data-driven predictive models, leading to significant improvements in energy efficiency [4].

The integration of AI into smart grids represents a pivotal advancement in modern energy infrastructure, aimed at enhancing efficiency, reliability, and sustainability. AI techniques, including deep learning and ML, have been widely applied to optimize energy generation, distribution, and consumption processes within smart grid frameworks [5]. These technologies enable predictive maintenance, real-time load balancing, and improved demand-side management, thereby reducing operational costs and environmental impact [6].

AI has become central to smart grid optimization in energy systems, yet several critical challenges hinder its full-scale deployment. One of the foremost issues is model interpretability, where many deep-learning and black-box AI models lack transparency, making it difficult for grid operators to trust and act on AI-driven recommendations [7]. Additionally, data heterogeneity and quality remain significant hurdles; smart grids produce high-frequency, multivariate, and often noisy data streams that traditional AI systems struggle to process effectively without extensive preprocessing and standardization [8]. Another core challenge is scalability and latency, especially in real-time applications, where edge devices may lack the computational power to run sophisticated AI models, leading to delays in fault detection or control decisions [9].

Large language models (LLMs) are advanced deep-learning architectures, typically based on transformer networks, which are trained on vast corpora of textual data to perform a wide array of natural language processing tasks such as text generation, summarization, and contextual understanding [10]. In recent years, LLMs have gained significant attention in the energy systems domain due to their capacity to automate complex modeling tasks, support decision-making processes, and enhance energy efficiency. Their application in building energy systems includes automating simulation input generation, fault detection, and real-time optimization [11].

Real-time optimization and fault detection in modern power and energy systems require intelligent models capable of reasoning, abstraction, and contextual understanding capabilities that traditional AI and ML methods often lack.

One essential factor necessitating LLM integration is the complexity and semantic diversity of smart grid data. These data include sensor signals, historical outage logs, weather inputs, and maintenance schedules. LLMs can unify and interpret these diverse data sources with context-aware language understanding [12]. Moreover, adaptability to dynamic grid conditions such as load fluctuations or equipment failures requires continuous learning and real-time prompt engineering. These both enable through few-shot and zero-shot capabilities LLMs can significantly enhance fault detection and predictive maintenance in smart grids by enabling intelligent

analysis of heterogeneous and unstructured data in real time. Furthermore, LLMs can support adversarially robust fault detection, enhancing grid resilience against cyber threats by identifying abnormal patterns hidden in operational narratives or system feedbacks [13].

1.1 | Scope of the Study

The scope of this study is to critically examine the integration of LLMs within the context of smart grid management, with a particular focus on fault detection, predictive maintenance, and real-time operational optimization. This research seeks to address the challenges posed by the complexity and heterogeneity of data generated within smart grids, including sensor readings, weather data, historical maintenance logs, and operator narratives. The study is aimed at exploring the capacity of LLMs to process and synthesize these multimodal data to improve grid efficiency, resilience, and decision-making processes. Furthermore, the study examines the potential of LLMs to improve the interpretability and transparency of AI-driven decisions in fault diagnosis. This area currently poses a significant challenge in existing ML models due to the inherent opacity of deep-learning systems.

1.2 | Literature Selection

This review adopts a structured narrative approach to analyze recent advancements in the application of LLMs in smart grid management. Relevant literature was collected from major scientific databases including IEEE Xplore, Scopus, Web of Science, and Google Scholar. The search was conducted using combinations of keywords such as “large language models,” “smart grid,” “fault detection,” “predictive maintenance,” “real-time optimization,” and “AI-based energy systems.” Peer-reviewed journal articles and high-quality conference papers published primarily between 2021 and 2025 were considered. Studies were included if they addressed AI- or LLM-based methods applied to power systems, smart grids, or energy optimization. Articles not related to energy systems, non-English publications, and non-technical opinion pieces were excluded. The selection process involved an initial screening of titles and abstracts, followed by full-text review to ensure relevance and technical contribution. The final set of studies was synthesized thematically to identify trends, challenges, and emerging integration pathways for LLMs in smart grid management. This approach ensures transparency while aligning with the exploratory and conceptual nature of the review.

1.3 | Contribution of the Study

The available literature affirms that innumerable surveys have been conducted to study the implementation of AI and ML methods to manage a smart grid. However, most of these reviews focus on algorithmic comparisons or single case studies such as load forecasting, fault classification, or energy trading. Such studies typically consider AI models as discrete numerical forecasts and only give scarce synthesis as to how to integrate them system-wide, interpretability, and semantic

reasoning. In contrast, the current review is aimed at achieving a unique academic contribution by redefining LLMs as semantic and supervisory intelligence layers integrated into smart grid designs. Instead of a simple list of algorithms, the given paper will synthesize the recent literature into a single conceptual framework that can map smart grid operational functions to the capabilities of LLM, such as fault detection, predictive maintenance, real-time optimization, self-healing, and cybersecurity, with the capabilities of the latter, such as contextual reasoning, multimodal data fusion, explainability, and uncertainty awareness, being mapped onto them. Furthermore, the review systematically classifies the current literature into the paradigms of LLM integration such as the LLMs with DTs, knowledge graphs (KGs), Bayesian learning, and reinforcement learning, outlining the complementary role of LLMs, as opposed to replacing the traditional ML models. Through this, the paper moves beyond literature aggregation and provides a structured analytical perspective that explains the realistic role, constraints, and deployment pathways of LLMs in future smart grids management systems.

1.4 | Organization of the Paper

The remainder of the article is summarized as follows: Section 2 reviews existing studies on AI applications in smart grids, emphasizing the limitations of traditional ML approaches and outlining the potential of LLMs to address these shortcomings. The next section, Section 3, is very important and examines the key challenges associated with integrating LLMs into smart grid systems. In Section 4, the paper discusses critical real-time grid optimization factors, highlighting key power and energy system functions where LLMs can enhance fault detection and operational efficiency. The next section, which is Section 5, shows the LLMs integration to improve fault detection and predictive maintenance in smart grids. Section 6 addresses the remaining integration challenges affecting accurate fault detection and real-time operational optimization. The next section, which is Section 7, represents the critical comparison of LLMs-based and traditional AI approaches in smart grids. Section 8 is the research gaps and future directions section, and at the last, there is Section 9 for conclusion. Table 1 presents the list of abbreviations used throughout the paper. Figure 1 shows the structure of the paper.

2 | From Traditional ML to LLMs: Evolution, Challenges, and Pathways in Smart Grid Management

2.1 | The Evolution of AI and ML in Power Systems: A Historical Perspective on Intelligent Grid Advancements

AI and ML have progressively become essential tools in modern power systems, enhancing performance across diverse tasks such as energy forecasting, grid stability, fault diagnosis, and asset management [14]. Initially applied in supervised forms for load prediction and fault classification, AI methods now encompass advanced models capable of managing large-scale, nonlinear dynamics in real-time environments [15]. In power

TABLE 1 | List of abbreviations.

Abbreviation	Definition
LLMs	Large language models
AI	Artificial intelligence
IoT	Internet of things
ML	Machine learning
SCADA	Supervisory control and data acquisition
DERs	Distributed energy resources
DERMS	Distributed energy resource management systems
RAG	Retrieval-augmented generation
BNNS	Bayesian neural networks
LSTM	Long short-term memory
GAN	Generative adversarial network
DRL	Deep reinforcement learning
BERT	Bidirectional encoder representations from transformers
T5	Text-to-text transfer transformer
GPT	Generative pretrained transformer

electronics, AI-driven control strategies have significantly improved the design and efficiency of power converters, particularly in renewable energy and electric drive applications. The use of AI in predictive maintenance and real-time diagnostics has also gained momentum, reduced downtime, and extended the lifespan of critical components [16]. Simultaneously, ML techniques like artificial neural networks, support vector machines, and deep-learning models have been widely adopted for forecasting energy consumption and integrating distributed energy resources (DERs), especially under volatile demand–supply conditions [17]. Comprehensive bibliometric analyses show a significant growth in AI–ML-related publications and patents in the energy sector over the last 5 years [18].

The purpose of this study [19] is to improve energy efficiency by using smart grids that are made possible by the IoT. These grids have both good and bad points. This article talks about how advanced buildings use the IoT. Use IoT devices to test smart grids. ML and SGs could help make energy use more efficient. We talk about the pieces and characteristics of the SG.

Furthermore, systematic reviews underscore the role of AI in modern energy management systems, with applications spanning smart grids, demand forecasting, energy trading, and resilience modeling [20]. Overall, the integration of AI–ML into power systems has evolved from isolated analytical tasks to becoming a cornerstone of intelligent, autonomous, and adaptive energy infrastructure. This study [21] investigates the diverse threats and vulnerabilities that may compromise the security of essential, advanced grid network components. Then, we offer security solutions that use a number of different methods to protect against these risks. We also give tips

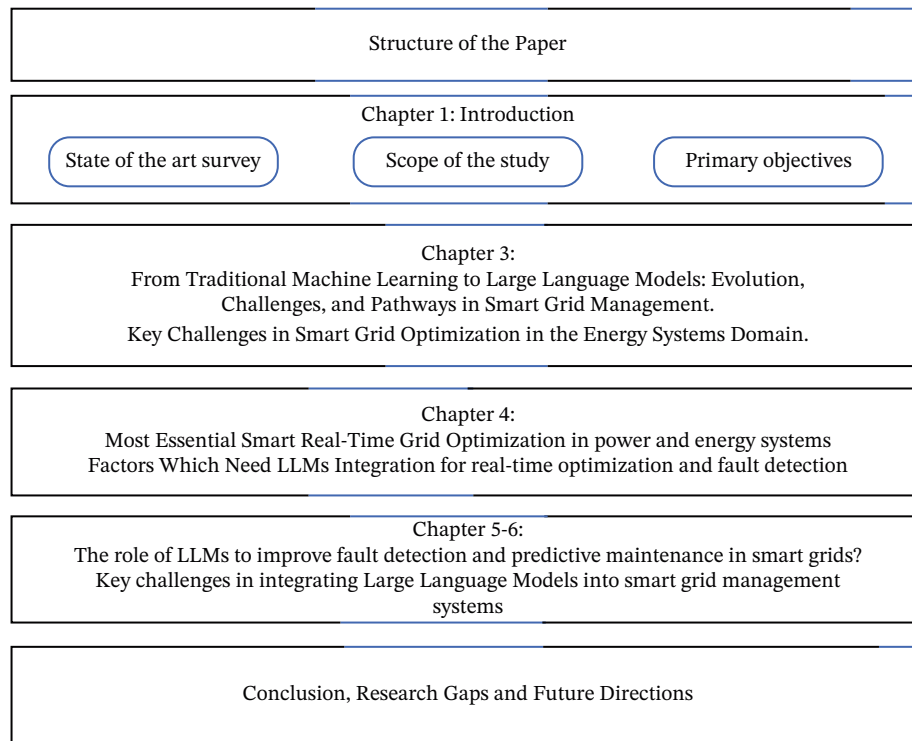


FIGURE 1 | Structure of the paper.

on how to make these three types of cyberattacks less likely to happen.

2.2 | Limitations of Traditional ML Techniques and Rise of LLMs

Traditional ML techniques, though foundational in energy systems, face persistent limitations in scalability, adaptability, and interpretability. Most models require extensive and domain-specific feature engineering, which hampers generalizability and increases implementation costs in dynamic environments like smart grids [22].

Scalability remains a challenge due to the high computational overhead needed to retrain or tune traditional models when exposed to evolving data streams or system configurations. Furthermore, classical ML models often function as black boxes lacking transparency and explainability in critical infrastructure, which raises reliability and ethical concerns in automated energy decision-making [23]. These shortcomings have paved the way for a paradigm shift with the rise of LLMs, which provide advanced contextual reasoning and multimodal capabilities to overcome traditional bottlenecks. LLMs have demonstrated the ability to significantly improve time-series prediction accuracy for electric and heat loads in integrated energy systems by eliminating the need for manual feature selection as shown in Figure 2 [24]. In distributed energy management, LLMs can integrate domain-specific safety constraints into reinforcement learning workflows, leading to enhanced performance and reduced operator intervention [25]. LLMs also offer self-adaptive scheduling for distributed renewable energy manufacturing,

leveraging experiential learning to make near real-time decisions that improve energy efficiency and reduce operational delays [26].

2.3 | Smart Grid Management and Data Complexity

2.3.1 | Multilayered Nature of Smart Grids: Sensors, IoT, SCADA Systems, and DERs

A smart grid is a technologically advanced and digitally enhanced electricity network that integrates communication, control, and automation capabilities into traditional grid infrastructure to enable real-time monitoring, bidirectional power flow, and optimized energy management [26]. This architecture is inherently multilayered, beginning with the sensor layer, which includes smart meters, voltage and current sensors, and phasor measurement units (PMUs) to continuously collect grid parameters for real-time analytics [27]. On top of this lies the IoT layer, which facilitates the transmission and processing of sensed data over networks using cloud-enabled or edge-based architectures [28]. The SCADA (supervisory control and data acquisition) system forms the operational control core, interfacing with sensors and actuators to ensure safe and efficient dispatch of electricity by offering centralized command and visibility [29]. Integrated within this architecture are DERs such as solar panels, wind turbines, electric vehicles, and battery storage units, whose decentralized nature introduces volatility but also enhances flexibility and resilience in grid operations [30]. Their coordinated integration requires intelligent management systems like DERMS (Distributed Energy Resource Management Systems), which communicate with SCADA and IoT nodes to balance loads and maintain voltage and

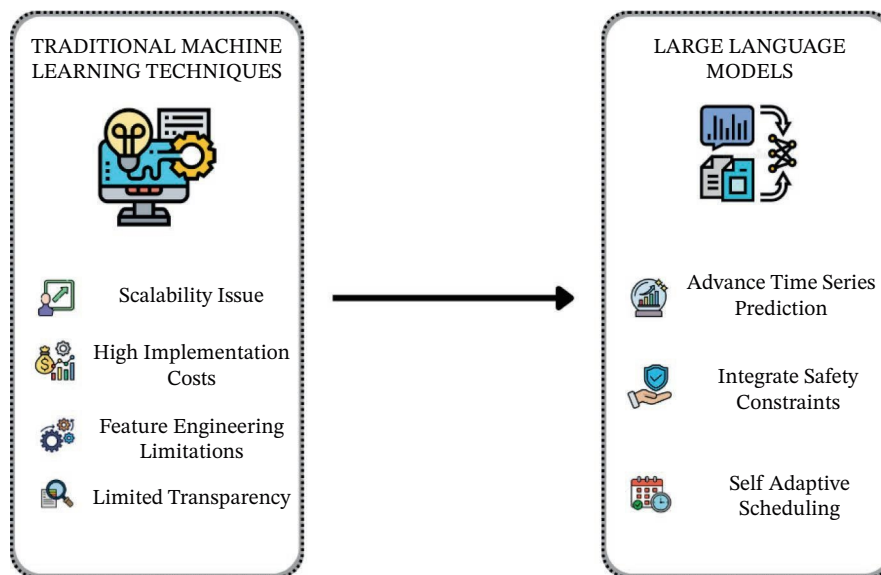


FIGURE 2 | Rise of large language models.

frequency stability [31]. The synergy of these layers enables smart grids to adapt in real time to fluctuations in demand, generation, and potential faults, thereby promoting a more secure, sustainable, and efficient energy ecosystem [32].

2.3.2 | Heterogeneity and Real-Time Demands of Smart Grid Data

Smart grid systems are inherently characterized by high data heterogeneity and stringent real-time processing demands due to their integration of diverse cyber-physical components. The data generated across smart grids are multisource and varies in structure, granularity, and temporal resolution, ranging from voltage phasors and power loads to customer usage patterns and DER statuses [33]. This heterogeneity is further compounded by the dynamic data access requirements introduced by the proliferation of IoT sensors and edge devices, which necessitate real-time fusion and analysis for reliable grid operation. The real-time constraints of smart grid analytics are further complicated by the need for secure and traceable data flow, which blockchain-based architectures have started to address by improving data integrity and reducing delays in provenance tracking. These demands call for robust, scalable, and intelligent data processing frameworks capable of maintaining low-latency performance even under high-volume, high-velocity data conditions, as demonstrated in recent convolutional and recurrent neural models for semantic data fusion [34]. Consequently, managing smart grid data involves addressing not just scale and speed but also context-awareness, interoperability, and resilience to failures in data transmission or processing pipelines [35]. These evolving requirements underscore the importance of intelligent, adaptive systems designed to handle real-time, heterogeneous, and decentralized data in future-proof smart grid architectures.

2.4 | Bridging LLMs and Smart Grids: Architectural Advances and Emerging Integration Pathways

The evolution of LLMs has been marked by foundational innovations in architecture and novel training strategies that support

their widespread application across domains. GPT, BERT, T5, and LLaMA represent archetypes optimized for different tasks ranging from generative modeling to sentence classification. While LLMs continue to perform exceptionally on unstructured text, advances in their application to structured and sequential data are ongoing, driven by research into hybrid models, dynamic attention, and efficient fine-tuning. As models become more efficient and ethically aligned, LLMs will further redefine the frontier of machine intelligence.

2.4.1 | Technical Foundations of LLMs

LLMs are built upon the transformative transformer architecture, a deep-learning structure that revolutionized sequence modeling by introducing the self-attention mechanism, allowing models to weigh relationships between words in a sequence dynamically and in parallel. Unlike recurrent models, which process inputs sequentially, transformers leverage self-attention and positional encoding to capture both local and global dependencies in a single pass, dramatically improving scalability and performance [36]. Recent architectural enhancements include dynamic attention and rotary positional embeddings (RoPE), which extend the model's ability to handle long sequences and improve contextual integrity during inference [37]. These innovations have led to the emergence of multimodal transformers, capable of processing text, vision, and audio simultaneously, a crucial development toward general-purpose AI [38].

Robustness and interpretability are becoming critical research frontiers. Techniques like ProTransformer, a plug-and-play enhancement layer, have shown improved resistance to adversarial attacks across BERT, T5, and LLaMA without retraining, underscoring the growing need for more secure and resilient LLMs in high-stakes applications [38]. Figure 3 shows the leading LLMs, and Table 2 presents a comparative analysis of traditional ML across smart grid system layers. Additionally, parameter-efficient fine-tuning techniques, such as LoRA and adapters, now enable effective task adaptation with minimal resource overhead [42].

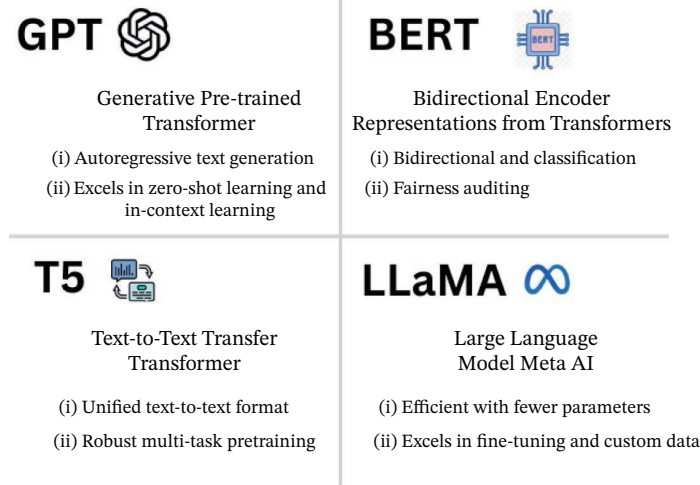


FIGURE 3 | Leading LLM architectures.

2.5 | Leading LLM Architectures: GPT, BERT, T5, and LLaMA

LLMs have recently been explored for smart grid applications due to their ability to process heterogeneous data sources such as SCADA logs, maintenance records, fault reports, and operational manuals. Unlike traditional ML models that rely primarily on numerical inputs, LLMs enable semantic reasoning across structured and unstructured grid data, supporting fault diagnosis, predictive maintenance, and supervisory decision-making in energy systems [8].

2.5.1 | GPT-Based Architectures for Fault Diagnosis and Reasoning

GPT-based architectures have been adapted for power systems primarily as high-level reasoning and decision-support tools. Fine-tuned GPT models demonstrate strong performance in interpreting fault scenarios by reasoning over historical outage records and real-time sensor descriptions, providing explainable diagnostic insights to grid operators. However, due to latency and computational overhead, these models are mainly deployed in supervisory roles rather than real-time protection systems [43].

2.5.2 | BERT-Based Architectures for Smart Grid Load Analysis and Monitoring

BERT-based architectures have been explicitly adapted for smart grid applications that involve sequential energy data analysis and contextual feature extraction. In particular, BERT models have been fine-tuned for transformer load forecasting and smart grid monitoring tasks by learning temporal and contextual dependencies from historical load datasets. These adaptations demonstrate BERT's effectiveness in extracting high-level features from energy consumption patterns, achieving improved forecasting accuracy compared with conventional deep-learning models, while maintaining suitability for operational grid analytics [44].

2.5.3 | T5-Based Architectures for Maintenance and Decision Support

T5-based architectures reformulate diverse grid management tasks into a unified text-to-text framework, enabling their application to fault summarization, maintenance planning, and operator decision support. In energy systems, T5-style models are fine-tuned to map fault descriptions to corrective actions, supporting automated documentation and maintenance workflows that enhance grid operational efficiency [45].

2.5.4 | LLaMA-Based Architectures for Efficient Domain Fine-Tuning

LLaMA-based models have gained attention in smart grid research due to their parameter efficiency and suitability for domain-specific fine-tuning. When integrated with DTs and knowledge-driven frameworks, LLaMA architectures support interpretable reasoning and scalable deployment for fault analysis and grid operation tasks. Their modular design enables tighter coupling with physical grid models, improving reliability in safety-critical environments [46].

2.6 | Bridging LLM Architectures and Smart Grid Applications

The above section introduces the general architectures of LLMs; their effective application to smart grid problems requires careful adaptation to structured, time-series, and multimodal grid data. Unlike natural language inputs, smart grid data consist primarily of high-frequency telemetry streams, event logs, and heterogeneous sensor measurements [47]. To bridge this gap, recent studies represent time-series grid data through tokenization strategies such as window-based segmentation, symbolic encoding of state transitions, or embedding numerical sequences into latent representations that can be processed by LLMs. Furthermore, multimodal fusion techniques play a critical role in enabling LLMs to reason

TABLE 2 | Comparative analysis of traditional ML and across smart grid system layers.

Smart grid layer/AI function	Limitations of traditional ML	Applications in energy systems	Focus area	Future work	Ref
Sensor layer	Limited by the need for extensive feature engineering. Struggles with scalability and real-time adaptability.	Real-time monitoring of grid parameters, such as voltage and current, for proactive fault detection and optimization.	Real-time data extraction and fault detection	Integrating multimodal sensor networks with LLMs for enhanced predictive capabilities.	[39]
IoT layer (communication and networking)	Traditional ML models struggle to handle heterogeneous, multisource data from IoT devices. High latency in decision-making.	Optimizing communication protocols and enhancing data flow between distributed devices like smart meters and sensors.	Data transmission and integration	Developing real-time encryption protocols for secure IoT-based communication.	[5]
SCADA system (operational control)	Traditional ML systems often lack transparency and interpretability, raising concerns in safety-critical decision-making.	Centralized monitoring and control of grid operations to ensure optimal energy distribution and real-time failure detection.	Grid operation and control	Enhancing real-time SCADA performance through hybrid AI systems combining traditional and LLM-based models.	[4]
Distributed energy resources (DERs)	Traditional ML struggles with the dynamic nature and scalability challenges of managing DERs, particularly in real-time.	Coordinating renewable energy sources (solar and wind) with grid needs for efficient energy distribution and load balancing.	DER management and optimization	Exploring LLMs for enhancing the dynamic control of DERs and improving grid response times.	[17]
Fault diagnosis	Classical ML often requires retraining and manual tuning to adapt to new failure modes or conditions.	Identifying and diagnosing faults in the grid, enabling predictive maintenance and minimizing downtime.	Fault detection and mitigation	Implementing explainable AI techniques to enhance interpretability in fault detection models.	[22]
Energy forecasting	Traditional ML models struggle to handle nonlinear and complex dynamics, leading to inaccuracies in long-term forecasts.	Forecasting energy consumption, generation, and demand-supply fluctuations to optimize grid stability and resource allocation.	Energy consumption prediction	Developing hybrid AI models combining LLMs and time-series analysis for better long-term predictions.	[40]

(Continues)

TABLE 2 | (CONTINUED)

Smart grid layer/AI function	Limitations of traditional ML	Applications in energy systems	Focus area	Future work	Ref
Load balancing and optimization	Traditional ML approaches face difficulties in optimizing energy distribution due to computational overhead and system complexity.	Ensuring optimal energy distribution across diverse consumer groups and grid layers, reducing operational delays.	Real-time optimization and load balancing	Developing edge-based AI systems for faster, localized decision-making in grid optimization.	[41]
Energy trading	Traditional ML models often lack adaptability to market changes and dynamic conditions, leading to suboptimal trading decisions.	Enhancing market predictions and trading strategies to optimize energy buying and selling based on real-time data.	Energy market optimization	Researching the use of real-time LLMs in market price prediction and energy trading automation.	[25]
Energy storage management	Traditional ML models have limitations in managing the complexities of energy storage systems, particularly about charge and discharge cycles.	Managing battery storage systems for peak demand periods, balancing load during intermittent renewable generation.	Energy storage and optimization	Implementing adaptive learning models to enhance real-time decision-making for energy storage systems.	[20]
Grid resilience modeling	Traditional ML lacks the ability to effectively model and predict grid resilience under extreme or unforeseen conditions.	Improving grid robustness and resilience in the face of natural disasters, equipment failures, or sudden demand surges.	Resilience analysis	Developing hybrid AI models for better predictive resilience modeling and failure prevention.	[17]

across diverse data sources, including SCADA measurements, phasor data, maintenance records, and operator annotations [48]. These approaches typically combine numerical encoders, graph-based representations of grid topology, and textual embeddings within a unified learning framework. By integrating structured time-series data with semantic and contextual information, LLMs can support higher-level reasoning tasks such as fault interpretation, predictive maintenance, and decision support [49].

Although Table 2 outlines the limitations of traditional ML and the potential roles of LLMs in smart grid systems, different LLM-based approaches exhibit distinct strengths and limitations when applied to the same task. For fault diagnosis, GAN-LLM hybrid models are preferable in scenarios where historical fault data are scarce, imbalanced, or subject to adversarial manipulation, as generative models can synthesize representative fault and attack samples to improve robustness and generalization. In contrast, Bayesian LLM-based approaches are more suitable for safety-critical fault detection, where uncertainty quantification and confidence-aware decision-making are required to reduce false alarms and

operational risk. Interpretable LLM frameworks integrated with belief rule bases (BRBs) offer greater transparency and traceability, making them preferable in operational environments where operator trust and explainability are mandatory. These tradeoffs demonstrate that no single LLM-based approach is universally optimal for fault detection, and the choice of method should depend on data availability, reliability requirements, interpretability needs, and real-time operational constraints.

3 | Key Challenges in Smart Grid Optimization in the Energy Systems Domain

Despite rapid progress in AI and LLMs, smart grid optimization within the power and energy systems domain faces 10 critical challenges that hinder real-world deployment. Scalability remains a key issue, as most AI models struggle to manage expansive and complex grids efficiently [50]. Explainability is equally pressing, as operators and stakeholders require transparent decision-making, but most AI models lack interpretability [51]. Cybersecurity vulnerabilities are exacerbated

as digital infrastructure increases, exposing smart grids to sophisticated cyberattacks [52]. Bias in AI models, often stemming from unbalanced training data, can lead to unfair energy distribution decisions [53]. Real-time data processing challenges also persist, with AI systems needing to process heterogeneous data streams at low latency for responsive control [54]. Additionally, integration complexity among AI, IoT, and legacy systems often delays implementation, whereas AI's energy consumption creates unintended loads on the grid itself [55]. Data privacy is a growing concern, especially with granular user-level consumption data being used in analytics [56]. Figure 4 shows the key challenges in smart grid optimization, and Table 3 represents the AI challenges in smart grid optimizations. Model generalizability also proves difficult, as AI trained in one grid context may not perform well in another [58]. Finally, interdisciplinary gaps between AI developers and domain experts remain a major bottleneck to practical application and adoption [59]. Bridging these multifaceted challenges will require integrated frameworks, cross-domain collaboration, and AI models tailored for operational transparency and domain-specific demands.

4 | Most Essential Smart Real-Time Grid Optimization in Power and Energy Systems Factors Which Need LLMs Integration for Real-Time Optimization and Fault Detection

4.1 | Fault Detection and Classification

Fault detection plays a foundational role in smart grid operation, ensuring grid stability by quickly identifying anomalies or equipment failures that may cause blackouts. In modern grids, real-time fault classification is crucial due to the proliferation of sensors and distributed devices, which generate complex, heterogeneous data streams; Figure 5 shows the most essential smart grid real-time optimization factors. Traditional rule-based systems or shallow ML models often fail to generalize in dynamic conditions or process large-scale telemetry efficiently [60]. LLMs are urgently needed here to semantically interpret both structured sensor data and unstructured logs such as textual operator reports providing interpretable, real-time fault classification across multimodal data environments.

4.2 | Demand Forecasting and Load Balancing

Real-time demand forecasting is essential to align energy generation with consumer needs, avoid blackouts, and ensure operational cost efficiency. Forecasting energy demand plays a crucial role in optimizing power generation and maintaining a stable energy supply. Traditional forecasting methods often struggle with uncertainty and complexity, which is why advanced techniques like LSTM and GRU models are increasingly used for accurate forecasting [61]. However, conventional statistical models or even classical ML methods lack adaptability when facing sudden demand spikes, irregular user behavior, or weather-influenced usage patterns. Traditional models may fail to adapt to sudden shifts in demand or weather-related anomalies, which is why the latest research focuses on dynamic systems capable of real-time adaptability [62]. Forecasting must process both structured data (historical loads) and contextual cues (user feedback and weather alerts), which LLMs are adept at synthesizing. Long short-term memory (LSTM) networks and other ML models have shown remarkable success in integrating diverse datasets, improving forecast accuracy [63].

4.3 | Renewable Integration Management

The intermittent and stochastic nature of renewables like wind and solar makes their integration into smart grids inherently unstable without continuous optimization [64]. Solar generation may fluctuate due to cloud cover, and wind turbines can vary dramatically even within minutes. Traditional forecasting or integration methods often underperform due to delayed reaction times or limited granularity [65]. LLMs can interpret semantic weather reports, maintenance logs, and grid feedback simultaneously to suggest immediate integration strategies. Their integration enables dynamic grid balancing and quicker curtailment or ramping of sources [66].

4.4 | DER Coordination

As prosumers and microgrids become more prevalent, smart grid coordination must evolve to handle decentralized, bidirectional energy flow. The emergence of prosumers and the

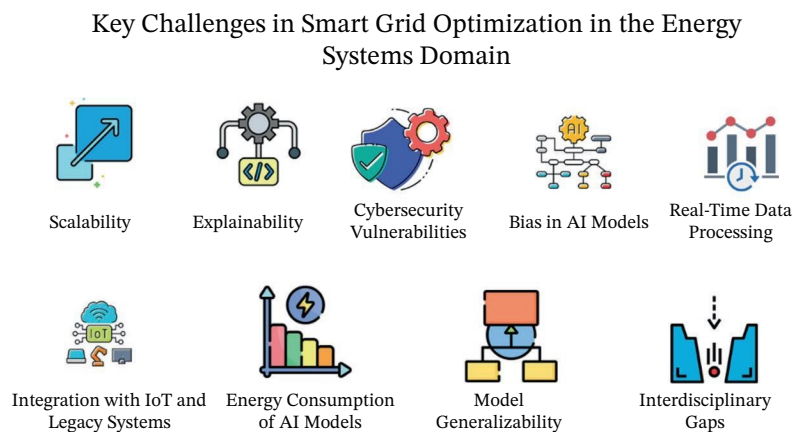


FIGURE 4 | Key challenges in smart grid optimization.

TABLE 3 | AI challenges in smart grid optimizations.

Challenge	The role of the challenge in smart grid management	Description of the challenge	Disadvantage	Future direction	Ref.
Scalability	AI models struggle with managing complex, large-scale grids.	AI models often cannot scale effectively to handle the large, diverse, and dynamic data from smart grids, which can lead to inefficiencies in grid management.	Inefficiency in real-time energy distribution and grid operation.	Developing scalable AI models capable of handling large datasets and providing real-time decision-making.	[50]
Explainability	Lack of transparency in AI decision-making.	AI models are often black-box models, making it difficult for operators to understand their decision-making processes, which is critical for trust and transparency in grid operations.	Lack of trust and inability to justify decisions in critical systems.	Research into explainable artificial intelligence (XAI) approaches for smart grid systems to improve transparency and accountability.	[51]
Cybersecurity vulnerabilities	Protection of grid infrastructure from cyber threats.	Increased reliance on digital technologies and networks introduces vulnerabilities in smart grids, exposing them to cyberattacks such as data breaches, denial-of-service (DoS), and malware.	Risk of large-scale disruptions, data loss, and unauthorized access.	Developing robust cybersecurity protocols and integrating AI-powered security systems to detect and prevent cyberattacks.	[52]
Bias in AI models	Ensuring fairness in AI-based decision-making.	AI models trained on biased datasets may perpetuate discrimination, resulting in unfair energy distribution and resource allocation.	Unfair distribution of resources and potential inequalities in service provision.	Implementing bias mitigation techniques and enhancing training data to ensure fairness and inclusivity in energy distribution decisions.	[53]
Real-time data processing	Processing dynamic and heterogeneous data efficiently.	AI systems must process large amounts of real-time data from multiple sources, which may be inconsistent or incomplete, to make quick decisions.	High latency and data inconsistency lead to delays in real-time grid management.	Developing real-time AI data processing systems capable of handling heterogeneous data streams efficiently.	[54]

(Continues)

TABLE 3 | (CONTINUED)

Challenge	The role of the challenge in smart grid management	Description of the challenge	Disadvantage	Future direction	Ref.
Integration with IoT and legacy systems	Combining new AI systems with existing infrastructure.	Integrating AI with existing smart grid infrastructure, such as IoT devices and legacy systems, can be complex and costly due to incompatibility and outdated technologies.	High integration costs and prolonged implementation timelines.	Focusing on the development of hybrid systems that enable seamless integration between modern AI technologies, IoT, and legacy systems.	[57]
Energy consumption of AI models	Balancing AI-driven efficiency and AI's power consumption.	AI systems themselves consume significant computational power, adding additional energy burdens to already strained grid resources.	Increased load on the grid, especially during high demand periods.	Research into energy-efficient AI models and hardware solutions designed for low power consumption without sacrificing performance.	[55]
Data privacy	Ensuring secure handling of user data.	AI systems in smart grids often require detailed user consumption data for analysis, raising concerns over data privacy and potential misuse.	Risk of privacy breaches and loss of public trust in AI systems.	Strengthening data privacy protocols using encryption, differential privacy, and federated learning methods.	[56]
Model generalizability	Ensuring AI models can adapt across diverse grid environments.	AI models trained in one grid environment may not generalize well to others, leading to suboptimal performance when applied to new or different grid configurations.	Poor performance and inefficiency when scaling models to different grid conditions.	Improving the generalization capabilities of AI models through transfer learning and domain adaptation techniques.	[58]
Interdisciplinary gaps	Collaboration between AI developers and domain experts.	Lack of communication and understanding between AI experts and energy domain specialists often leads to ineffective or incomplete AI solutions for smart grid applications.	Inability to develop fully optimized and effective AI models due to a lack of domain knowledge.	Promoting interdisciplinary collaboration between AI researchers and energy professionals to create solutions tailored to grid management needs.	[59]

rise of decentralized microgrids pose significant challenges for smart grid coordination, demanding new methods to handle bidirectional energy flow effectively [67]. Current optimization methods fall short in handling heterogeneous contractual, behavioral, and technical signals at scale. Traditional optimization methods are not sufficient for coordinating heterogeneous

signals from various DERs and microgrids, necessitating more advanced strategies [68]. LLMs fill this gap by parsing agreements, control logs, user preferences, and telemetry to orchestrate intelligent DER coordination. LLMs are able to process a variety of data types, including contracts, logs, and user preferences, to effectively manage DER coordination [69].

Essential Smart Real-Time Grid Optimization Factors Needing LLMs Integration

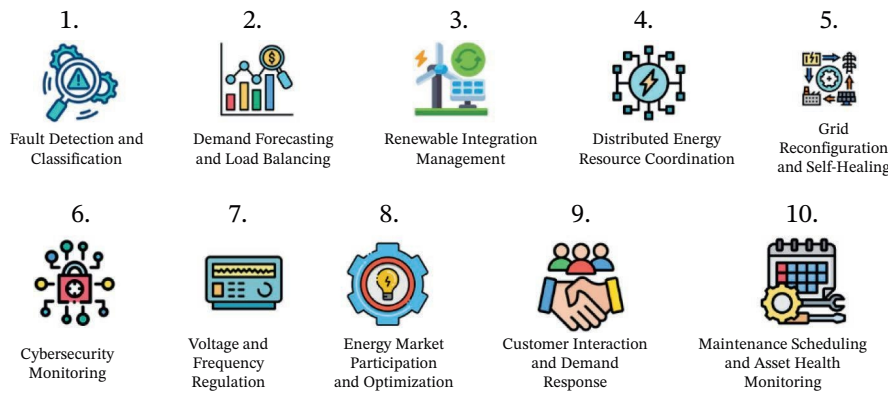


FIGURE 5 | Most essential smart grid real-time optimization factors.

4.5 | Grid Reconfiguration and Self-Healing

Smart grids need the ability to self-heal in response to faults by rerouting power through alternate paths. The need for self-healing in smart grids has led to the development of intelligent systems capable of dynamically reconfiguring the grid to maintain stability during faults [70]. However, the decision to reconfigure the grid topology must be made instantaneously, considering real-time loads, weather, and safety conditions. The complexity of real-time grid reconfiguration necessitates fast decision-making systems that account for multiple operational parameters, including load demands and environmental factors [71]. Traditional automation is limited to static rule-based switching, whereas LLMs can evaluate real-time logs, topology maps, and operator communications to suggest optimal paths during failure or overload. LLMs offer superior performance by analyzing diverse data types such as logs and operator communications to improve fault management and reconfiguration decisions [72].

4.6 | Cybersecurity Monitoring

Cybersecurity in smart grids is critical to prevent malicious attacks that could disrupt power delivery, manipulate control commands, or exploit data vulnerabilities. As grid infrastructures increasingly depend on digital communication and control systems, the risk of cyber threats grows exponentially [73]. Traditional rule-based intrusion detection systems (IDS) struggle to recognize novel or evolving threats and fail to analyze complex command or telemetry data. They often lack semantic understanding of attack patterns and real-time adaptability. LLMs can monitor unstructured logs, system alerts, and anomalous communication patterns by contextualizing text-based commands and logs in real time. This enhances intrusion detection and enables faster, automated mitigation [74].

4.7 | Voltage and Frequency Regulation

Maintaining voltage and frequency within allowable bounds is vital to grid stability. The need for stable voltage and frequency

regulation is critical in power systems, especially with the high integration of renewable energy sources and varying grid conditions [75]. Deviations may result in cascading outages or equipment damage, especially during high penetration of renewables or heavy load conditions [76]. Conventional control systems are slow in adapting to rapid changes, especially with variable renewable sources, and are not capable of integrating qualitative data like operator notes or weather anomalies. Traditional control systems struggle to quickly adapt to the dynamic behavior of renewable energy, creating challenges for maintaining stable grid operations [77]. LLMs enhance voltage and frequency control by synthesizing SCADA data, equipment status, and textual reports to produce real-time diagnostic insights and tuning recommendations. LLMs have proven effective at processing diverse data types and offering actionable insights for grid management [78].

4.8 | Energy Market Participation and Optimization

Modern energy markets require grid operators and consumers to make dynamic, time-sensitive decisions to buy, sell, or store energy efficiently [79]. Traditional optimization techniques cannot parse legal texts, contracts, or policy regulations in real time. Additionally, they cannot account for nonnumeric factors like social media sentiment or public policy announcements. LLMs interpret complex contracts, market news, and real-time bid strategies, offering strategic insights and price predictions that conventional models cannot provide [80].

4.9 | Maintenance Scheduling and Asset Health Monitoring

Predictive maintenance helps avoid costly breakdowns and ensures grid reliability by anticipating faults and scheduling timely interventions [81]. Conventional monitoring relies heavily on structured sensor data and fails to incorporate textual feedback from technicians, failure logs, or past service notes. LLMs excel in combining structured data with unstructured inputs like service reports and audio transcripts

to identify failure trends and propose optimal maintenance schedules [82].

4.10 | Customer Interaction and Demand Response

Demand response mechanisms involve engaging consumers to adapt their energy use based on grid conditions. Personalized communication boosts participation and improves peak load management [80]. Standard communication protocols are rigid, and most do not personalize alerts or incentives, leading to lower user engagement. LLMs can interpret user preferences, feedback, and natural language patterns to craft personalized messages and real-time alerts that significantly enhance user participation and responsiveness [83].

5 | LLMs Integration to Improve Fault Detection and Predictive Maintenance in Smart Grids?

LLMs enhance fault detection by interpreting heterogeneous data such as sensor streams, maintenance logs, and operator reports in real time. They enable predictive maintenance through contextual reasoning, identifying patterns and anomalies before equipment failure occurs. LLMs support adversarial-robust detection, improving resilience against cyberattacks and false data injection. By combining structured telemetry with unstructured narratives, LLMs provide interpretable and reliable diagnostic insights. Their integration reduces downtime, optimizes asset health monitoring, and strengthens overall grid reliability. Figure 6 below shows the integration of LLMs in smart grids to overcome different challenges, and Table 4 shows a summary of LLM-based solutions in smart grids.

5.1 | Adversarial-Robust LLMs for Fault Classification in Smart Grids

Adversarial-robust LLMs and deep-learning frameworks are increasingly being used to enhance fault classification and

predictive maintenance in smart grids by improving resilience against cyberattacks and data manipulation. The study by [13] introduces FaultGuard, a novel architecture that combines generative adversarial networks (GANs) with LLM-style classifiers to detect and classify faults in smart grids, particularly under adversarial input conditions. The system enhances robustness by simulating realistic fault and attack scenarios through the GAN, whereas the transformer-based classifier provides strong generalization across unseen data. Their model demonstrated a classification accuracy detection rate for adversarial inputs. The importance of adversarial robustness in fault detection pipelines is further validated by [84] who demonstrated how LLMs can detect false data injections when fused with telemetry analytics.

5.2 | Interpretable LLMs With BRBs

Interpretable LLMs empowered by BRBs are being increasingly adopted for fault detection and predictive maintenance in smart grids and other complex systems due to their ability to fuse expert knowledge with data-driven reasoning. The study of the authors [7] shows the adaptive interpretable belief rule base (AI-BRB) ensures interpretability during model optimization by adaptively updating the search space and introducing interpretability constraints, achieving a balance between accuracy and transparency in smart grid fault detection. The study of [85] echoed the need for interpretability in their GAN-fuzzy-LLM hybrid system, showing how integrating rule-based systems can improve both accuracy and trust. The techniques developed by [86] micro-belief rule base (MBRB) address the complexity of large rule structures through cautious conjunctive reasoning and an interpretable evolutionary optimization strategy to retain diagnostic clarity.

5.3 | DT Systems Powered by LLMs

Recent advancements in LLMs combined with DT systems have significantly enhanced fault detection and predictive maintenance in smart grids. One effective approach integrates

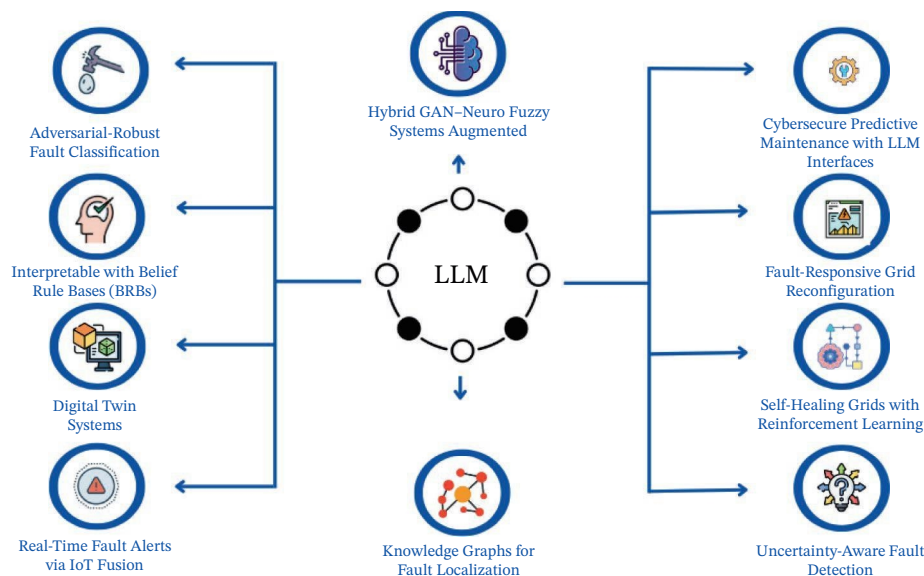


FIGURE 6 | Integration of LLMs in smart grid to overcome different challenges.

TABLE 4 | Summarizing table of LLM-based solutions in smart grids.

Important factor name	LLMs-based model used	Objective in energy efficiency in smart grids	Application domain	Effectiveness	Advantages	Disadvantages
Adversarial-robust fault classification	FaultGuard (GAN and LLM transformer)	Enhance fault classification under adversarial conditions	Cybersecurity, fault classification	High classification accuracy even under adversarial inputs	Robustness to false data injection, generalizes to unseen faults	Training GANs and transformers can be computationally intensive
Interpretable fault detection	AI-BRB	Transparent decision-making in predictive maintenance	Interpretable AI, expert systems	Balances interpretability with predictive accuracy	Human-readable logic, rule-base transparency	Scalability issues with large rule bases
Digital twin systems	RAG and knowledge graphs LLMs	Simulate grid behavior and enable scenario planning	Digital twin, simulation	Improves real-time decision-making and scenario simulation	Context-aware simulations, supports edge case analysis	Data-heavy; dependent on accurate digital modeling
Bayesian fault detection	Bayesian LLMs with BNNs	Uncertainty-aware predictions to reduce false alarms	Reliability analysis	High reliability through uncertainty quantification	Improved trust and interpretability of predictions	May introduce computational complexity
Self-healing grids	LLMs and reinforcement learning (IRL, DRL)	Autonomous fault handling and reconfiguration	Control systems, automation	Enables real-time reconfiguration and resilience	Autonomy, adaptability to dynamic conditions	Requires extensive training and tuning
IoT and LLM fusion	Transformer-based LLMs and IoT telemetry	Real-time anomaly detection and alerts	Sensor networks, fault monitoring	Improves fault detection latency and accuracy	Scalable, integrates structured/unstructured data	Sensitive to sensor quality and network reliability
Hybrid GAN-neuro fuzzy systems	GAN, fuzzy logic and LLM Interfaces	Interpretability and robustness in classification	Data augmentation, fault recognition	Improves accuracy with balanced datasets	Combines interpretability with data generation	Complex model integration and tuning
Grid reconfiguration	LLMs and logical structure	Adaptive fault rerouting and recovery	Grid topology management	Fast and intelligent switching during faults	Context-aware grid optimization	Transferability to large-scale systems can be complex
Cybersecure maintenance	LLM interfaces and ML IDS	Secure and private predictive maintenance	Cybersecurity, privacy	Detects and defends against cyber threats	Enhances trust and data protection	Overhead of federated learning and security layers
Knowledge graphs for fault localization	KG and prefix-tuned LLMs	Contextual and explainable fault localization	Fault diagnostics, semantic AI	High accuracy and traceability in fault paths	Semantic clarity and expert system integration	KG maintenance and expansion require domain effort

retrieval-augmented generation (RAG) with LLMs and KGs to create conversational DTs that assist in maintenance decision-making by interpreting complex asset data, as demonstrated in energy infrastructure [87]. The study of [88] emphasized the role of semantic fault linking in real-time simulations, supporting LLMs' role in dynamic modeling. In another study [89], the authors showed that topology-aware digital simulations, when enhanced with semantic interpretation, can lead to optimized reconfiguration decisions.

5.4 | Bayesian LLMs for Uncertainty-Aware Fault Detection

Bayesian LLMs and Bayesian deep-learning techniques are increasingly used for uncertainty-aware fault detection and predictive maintenance in smart grids. In particular, Bayesian neural networks (BNNs) enhance reliability by quantifying predictive uncertainty (e.g., predictive entropy and mutual information), which can reduce false alarms and improve robustness under noisy or adversarial conditions when applied to fault zone prediction and decision support [86]. Related work has extended uncertainty-aware reasoning by correlating telemetry anomalies with suspicious operational logs to strengthen confidence-aware maintenance decisions [87]. Moreover, uncertainty-guided data augmentation and sampling strategies (including GAN-based approaches) can improve fault-data diversity and, in turn, the accuracy and calibration of Bayesian models in imbalanced or sparse-fault settings.

5.5 | LLM-Enabled Self-Healing Grids With Reinforcement Learning

LLMs combined with reinforcement learning are enabling the development of self-healing smart grids that can autonomously detect faults, perform predictive maintenance, and restore service with minimal human intervention. A significant contribution comes [90] from inverse reinforcement learning (IRL) approaches, where systems learn optimal control strategies from historical nonfault data and adaptively detect and compensate for nonlinear actuator faults without relying on a complete model of the system dynamics. The study of [91] developed deep reinforcement learning (DRL), enabling cloud-edge collaborative fault detection systems that allocate computing resources in real-time to minimize detection delay and optimize throughput in smart grid networks.

5.6 | Real-Time Fault Alerts via IoT and LLM Fusion

Real-time fault alerts in smart grids through the fusion of IoT and LLM technologies are transforming predictive maintenance by enabling highly responsive, intelligent systems that detect anomalies and trigger proactive responses. The study of [92] developed a hybrid architecture combining IoT sensor telemetry with LLM-driven text interpretation to support early fault alerts in smart grids. Their system processes real-time sensor anomalies and parses unstructured maintenance logs using transformer-based models to detect fault precursors with improved

reliability. Complementary work by [93] proposed integrating adversarial training into IoT-LLM systems to reject spoofed telemetry and maintain detection fidelity even under cyberattack scenarios. The study of [94] developed a real-time fault prediction framework by integrating IoT sensor data with cost-sensitive logistic regression and SVM classifiers for smart charging piles, achieving improved early fault detection and cost-efficient preventive maintenance strategies.

5.7 | Hybrid GAN-Neuro Fuzzy Systems Augmented by LLMs

The study of developing hybrid GAN-neuro fuzzy systems augmented by LLMs for fault detection and predictive maintenance in smart grids has led to several promising methodologies that integrate interpretability with generative intelligence. The study of [95] applied GANs for sample balancing in fuzzy rule-based classifiers, improving fault recognition accuracy in wind turbine data. The GAN-fuzzy logic hybrids enhance interpretability in real-time AI systems, which aligns with the need for semantic clarity in power grid fault analysis. In a grid security context, the study of [96] developed a hybrid CNN-GAN model enhanced with fuzzy logic for detecting energy theft using smart meter data, leveraging GANs to synthetically balance attack vector datasets and fuzzy systems for interpretability in classification.

5.8 | LLMs for Fault-Responsive Grid Reconfiguration

The study of developing LLMs for fault-responsive grid reconfiguration has led to diverse frameworks that combine intelligent reasoning, optimization, and adaptive control to enhance smart grid resilience and fault recovery. The study of [97] emphasized the need for stability-aware adversarial models that preemptively reroute energy in compromised segments using transformer-like interpretability layers. The study of [98] developed a two-stage fault reconfiguration strategy for ship DC regional grids using a logical structure (LS) model and improved reinforcement learning, enabling fast, intelligent switching based on fault severity and prioritization, which can be translated to smart grid fault reconfiguration by leveraging LLMs for strategy interpretation and policy adjustment. The study of [93] proposed a transfer learning-based transient stability prediction framework under evolving grid topologies using ConvLSTM networks, which supports real-time reconfiguration by retraining models to new scenarios, an approach where LLMs can further enhance contextual understanding and inference efficiency.

5.9 | Cybersecure Predictive Maintenance With LLM Interfaces

The study of developing cybersecure predictive maintenance systems with LLM interfaces in smart grids has led to a fusion of advanced AI models with robust cybersecurity architectures to ensure both operational reliability and protection against evolving threats. The study of [99] developed a holistic smart grid management framework integrating ML-based diagnostics, prognostics, and optimal power flow with a dedicated

cybersecurity module, ensuring predictive maintenance is protected from data breaches and unauthorized control access. The study of [100] expanded this work by creating adversarially trained LLM and GAN systems to detect evasive malware in energy infrastructure; it also supported interpretable cybersecurity auditing via belief-rule bases enhanced with LLMs for grid telemetry data. The study of [101] introduced a ML-integrated cybersecurity framework for WSNs in smart grids using random forests and traffic analysis for intrusion detection, a method that can be paired with LLMs for contextual understanding and anomaly classification.

5.10 | LLM and KGs for Fault Localization

The study of developing LLMs combined with KGs for fault localization in smart grids has produced robust and interpretable frameworks that unify deep reasoning, semantic context, and system-specific knowledge for precise diagnostics. The study of [102] developed the To-FD-EKG framework, integrating a fault diagnosis event knowledge graph (FD-EKG) with LLMs acting as agents to generate interpretable fault localization reasoning paths and actions, reducing hallucinations and improving traceability in wind turbine diagnostics. The study of [103] designed the PF2RM model, a fault retrieval and recommendation framework that reconstructs the KG ontology to cluster fault entities, improving retrieval performance during cold-start conditions. Similarly, the study of [104] proposed a joint model embedding KGs into LLMs via prefix-tuning for industrial fault localization, achieving high accuracy by generating subgraphs and performing retrieval-augmented reasoning within domain-specific environments.

6 | Key Challenges in Integrating LLMs Into Smart Grid Management Systems, Particularly for Achieving Accurate Fault Detection and Real-Time Operational Optimization?

Integrating LLMs into smart grid management systems presents several critical challenges that must be addressed to ensure effective fault detection and real-time operational optimization. LLMs offer transformative potential for smart grid fault detection and real-time optimization; their integration is hindered by security vulnerabilities, data limitations, explainability concerns, computational constraints, and organizational readiness, all of which must be addressed holistically for successful deployment.

- One of the key issues is the interpretability and traceability of LLMs in fault diagnosis scenarios. The study of [105] introduces the To-FD-EKG framework and emphasizes that LLMs, despite their generative reasoning capabilities, often struggle with factual hallucinations and a lack of transparency. This problem is especially critical in safety-critical applications like smart grids where decisions must be explainable and trustworthy. Another significant challenge is the limited ability of LLMs to process real-time data streams efficiently. The study of [106] highlights the difficulty of applying transformer-based architectures, such as Vision Transformers (ViTs), to

high-frequency PMU data due to their computational inefficiency in real-time fault detection. Although ViTs outperform CNNs in accuracy, their high resource demands hinder practical deployment for continuous monitoring tasks.

- Additionally, it addresses the cybersecurity risks associated with integrating AI and IoT technologies in smart grid environments. As LLMs increasingly interface with sensor networks for predictive maintenance, they become potential attack surfaces. Ensuring secure data exchange and model robustness against adversarial inputs remains an unresolved issue, especially in large-scale deployment scenarios.
- Furthermore, real-time system adaptability is constrained by the limited learning ability of LLMs under dynamic grid conditions. The study by [107] points out that faults often change the character of grid behavior unpredictably, and while traditional nonintrusive monitoring methods can adapt via derived power streams, LLMs lack mechanisms to revise their internal knowledge representations on-the-fly. Lastly, the challenge of integrating LLMs into multimodal semantic systems is further addressed by [108], who developed a deep-learning-based fault diagnosis model that incorporates KGs. While this combination improves detection accuracy, the authors note the difficulty of aligning LLM outputs with structured ontologies, especially in environments with high semantic heterogeneity like smart grids.
- The challenges discussed in this section are directly connected to the LLM-based solutions above. For example, the interpretability challenge highlighted above is explicitly addressed by the interpretable LLM frameworks integrated with BRBs discussed in above sections, which combine expert knowledge with data-driven learning to provide transparent and traceable fault diagnosis. Similarly, the cybersecurity and adversarial robustness challenges identified in this section are mitigated by the GAN-LLM hybrid architectures presented in above sections, which improve fault classification reliability under adversarial and manipulated data conditions. Uncertainty-related challenges in safety-critical grid operations are addressed by the Bayesian LLM-based approaches discussed in the above sections, where uncertainty quantification supports confidence-aware decision-making and reduces operational risk. These linkages demonstrate that the challenges identified in this section serve as design drivers for the LLM-based solutions reviewed earlier, rather than standalone limitations.

7 | Critical Comparison of LLM-Based and Traditional AI Approaches in Smart Grids

According to the reviewed literature, standard ML and deep-learning models remain highly effective in structured and low-latency tasks of smart grid applications, including numerical load forecasting, voltage regulation, and fast fault classification based on sensor data. Their predictable nature, lower calculation costs, and suitability to deployable edges make them reliable with respect to real-time control cycles and safety-related functions. However, LLMs have significant advantages where heterogeneous and unstructured data such as maintenance logs, operator reports,

cybersecurity alerts, and semantic fault descriptions are involved. In the literature reviewed, the approaches to LLM are more effective than the traditional ones in contextual reasoning, multimodal data fusion, and high-level decision support, especially regarding predictive maintenance, fault interpretation, and supervisory grid management. Despite these advantages, there are significant disadvantages that arise with the methods of LLM. Their large computational and inference latency limits their use in real-time and edge-based systems, where general AI models are more feasible.

Furthermore, without KGs, DTs, or rule-based reasoning schemes to reduce them, LLMs are susceptible to hallucinations, which can provide incorrect fault interpretation or dangerous advice. Explainability is also a tradeoff. Although LLMs are capable of generating explanations that are comprehensible to humans, these explanations are not necessarily consistent with falsifiable causation. Comparatively, interpretable conventional methods, including BRBs and rule-based expert systems, provide better traceability and operator confidence concerning safety-critical grid systems. Moreover, because of the utilization of large volumes of operation data, the implementation of LLMs exacerbates the threats of cybersecurity and data privacy. Overall, it is possible to state that LLMs cannot replace conventional AI-based approaches in smart grid systems. Instead, the best architectures are hybrid, that is, they use LLMs as a supervisory and semantic reasoning layer that supplements traditional ML models that address real-time control and numerical optimization.

7.1 | Real-World Deployment Constraints of LLMs in Smart Grid Systems

Despite their strong conceptual potential, the practical deployment of LLMs in real-world smart grid environments faces several critical constraints. One of the most significant barriers is latency, as many smart grid operations, including fault isolation and protection control, require millisecond-level response times that current LLM inference pipelines cannot reliably meet, particularly in edge-based settings. Although cloud-based deployment can mitigate computational limitations, it introduces communication delays and reliability concerns that are unacceptable for safety-critical grid operations. Another major constraint is the computational and energy cost associated with LLM training and inference. Smart grid infrastructures are increasingly aimed at reducing operational energy overhead, yet LLMs require substantial computational resources, which may conflict with sustainability and cost-efficiency objectives. Additionally, integration with legacy infrastructure, such as SCADA systems, DERMS, and protection relays, presents significant engineering challenges due to interoperability, protocol incompatibility, and certification requirements.

Operational reliability and safety represent further deployment barriers. Unlike traditional control algorithms, LLMs exhibit probabilistic behavior and may generate inconsistent or nondeterministic outputs, raising concerns for their use in mission-critical decision-making. As a result, human-in-the-loop supervision and rule-based constraints remain necessary, limiting the level of automation that can be realistically achieved. Furthermore, regulatory, cybersecurity, and data governance constraints restrict the use of centralized data-driven models in many regions,

particularly where critical infrastructure data cannot be transmitted to external cloud platforms. These practical limitations indicate that, while LLMs offer valuable high-level reasoning and decision-support capabilities, their real-world deployment in smart grids is currently best suited to advisory, supervisory, and postevent analysis roles, rather than direct real-time control. Addressing these deployment constraints remains essential for translating conceptual LLM capabilities into operationally viable smart grid solutions.

8 | Research Gaps and Future Directions

8.1 | Domain-Specific Pretraining

The study of [109] emphasizes that while pretrained LLMs show promise in extracting insights from unstructured data, their effectiveness is significantly enhanced through domain-specific fine-tuning. The integration of LLMs in smart grids demands the development of large-scale, domain-specific corpora encompassing multilingual maintenance logs, incident reports, and control policies. The absence of such datasets currently limits the ability of LLMs to generalize across heterogeneous grid systems and regional infrastructures.

8.2 | Edge AI and Federated Learning

The study of [105] emphasizes a key research gap that federated AI algorithms can be deployed at the 5G edge for low-latency analytics; the scalability and real-time responsiveness of these systems remain constrained by limitations in communication overhead and edge node heterogeneity. This is particularly critical when integrating LLMs, which are computationally intensive, into resource-limited edge environments.

8.3 | Synthetic Data and Simulation

The study of [110] emphasizes the importance of dataset diversity for fault classification in power grids and proposes the use of auxiliary classifier generative adversarial networks (ACGANs) to synthetically balance imbalanced fault datasets. They demonstrate that the performance of ensemble models such as Extra Trees and Gradient Boosting improves significantly when trained on GAN-augmented data. The study of [111] confirms that synthetically enriched data distributions enhance model robustness, particularly in grid environments with sparse historical fault data. Auto encoders and oversampling enhance traditional classifiers, LLMs require structured, narratively enriched, and temporally grounded synthetic sequences none of which are addressed by existing data augmentation techniques.

8.4 | Multimodal and Cross-Disciplinary Synergies

The study of [112] introduces the concept of the LLM multiplex, emphasizing the importance of integrating multimodal and multitiresponsibility capabilities into model architectures to meet complex decision-making tasks across disciplines. Although focused

on healthcare, this framework is directly transferable to smart grid systems, where future architectures must incorporate visual inspections from drones (CV), sensor telemetry (IoT), control logic (optimization), and maintenance narratives (NLP). The lack of unified codesign models that combine these streams remains a major limitation. Future research should prioritize building interpretable, cross-disciplinary frameworks capable of reasoning over multimodal inputs for fault localization and grid reconfiguration.

9 | Conclusion

This review synthesized recent advances on the use of LLMs in smart grid management, with emphasis on fault detection, predictive maintenance, and real-time operational optimization. Beyond summarizing prior work, the manuscript positions LLMs as semantic and supervisory intelligence layers that complement—rather than replace—traditional ML and control approaches. The reviewed studies collectively indicate that LLMs are particularly valuable when grid intelligence must integrate heterogeneous information sources (e.g., telemetry streams, SCADA/event logs, maintenance records, and operator narratives) and translate them into operator-facing insights, explanations, and actionable recommendations.

At the same time, the literature indicates that real-world implementation readiness remains constrained by several factors. These include inference latency and compute cost (especially under real-time protection requirements), risks of unreliable or nonverifiable generated outputs without grounding and safeguards, interoperability challenges with legacy infrastructures, and strict data governance constraints in critical energy systems. Since smart grids are increasingly exposed to sophisticated threats, cybersecurity and adversarial robustness must be treated as first-order design requirements; LLM-enabled pipelines can expand the attack surface through data poisoning, prompt/feedback manipulation, and false-data injection if not properly secured. Consequently, near-term deployments are best framed as human-in-the-loop decision support with strong guardrails—such as retrieval- or knowledge-grounded reasoning, uncertainty-aware outputs, auditability, and defense-in-depth cybersecurity controls—rather than direct closed-loop control in safety-critical functions. Future research should prioritize domain-specific benchmarks, calibrated uncertainty evaluation, standardized adversarial testing protocols, and deployment architectures that satisfy operational and regulatory requirements for critical infrastructure.

Author Contributions

Wasim Ahmad and Tehseen Mazhar performed the original writing part, software, and methodology. Syed Faisal Abbas Shah and Tehseen Mazhar performed rewriting, investigation, design methodology, and conceptualization. Tariq Shahzad, Habib Hamam, and Wajahat Waheed performed the related work part and manage results and discussions. Tehseen Mazhar, Khmaies Ouahada, and Habib Hamam performed related work part and manage results and discussion. Tehseen Mazhar and Tariq Shahzad performed rewriting, design methodology, and visualization. Tariq Shahzad and Habib Hamam performed rewriting, design methodology, and visualization.

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The authors confirm that no generative artificial intelligence (AI) systems were used to produce the scientific content, technical analysis, results, or conclusions presented in this manuscript. AI-assisted tools were used solely for language editing and stylistic improvement, as well as for the graphical design and formatting of conceptual figures based on the authors' original ideas. The use of such tools was limited to editorial and visualization support and did not influence the scientific contributions of the work.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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